0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24

Input Matrix

w0	w1
w2	w3

Kernel

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Output Matrix

w0	0	0
w1	1	0
w0	1	1
w1	2	1
w0	2	2
w1	3	2
w0	3	3
w1	4	3
w2	5	0

For the output index 0:

0	**1**	2	3	4
5	**6**	7	8	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24

Computation

0*w0 + 1*w1 + 5*w2 + 6*w3

For the output index 1:

0	**1**	**2**	3	4
5	**6**	**7**	8	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24

Computation

1*w0 + 2*w1 + 6*w2 + 7*w3

For the output index 2:

0	1	**2**	**3**	4
5	6	**7**	**8**	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24

Computation

2*w0 + 3*w1 + 7*w2 + 8*w3

Connection map:

Output neurons

Input N.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	w0															
1	w1	w0														
2		w1	w0													
3			w1	w0												
4				w1												
5	w2				w0											
6	w3	w2			w1	w0										
7		w3	w2			w1	w0									
8			w3	w2			w1	w0								
9				w3				w1								
10					w2				w0							
11					w3	w2			w1	w0						
12						w3	w2			w1	w0					
13							w3	w2			w1	w0				
14								w3				w1				
15									w2				w0			
16									w3	w2			w1	w0		
17										w3	w2			w1	w0	
18											w3	w2			w1	w0
19												w3				w1

.....

For the output index 4:

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24

Computation

5*w0 + 6*w1 + 10*w2 + 11*w3

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20													w2			
21													w3	w2		
22														w3	w2	
23															w3	w2
24																w3